USN | | | | | | | | | | [22CSE101]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

ARTIFICIAL INTELLIGENCE

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A		**5 X 2=10**	
1.	**a.**	Define Intractability with an example.	L1	CO1	**02**
	b.	Write short note on PEAS.	L3	CO2	**02**
	c.	Define search algorithm.	L1	CO3	**02**
	d.	Illustrate a node in a search tree.	L3	CO4	**02**
	e.	Define knowledge base system with block diagram.	L1	CO5	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Discuss how AI become a major part in Industory.	L2	CO1	**05**
		OR			
	b.	Describe briefly any 5 states of the art AI systems.	L2	CO1	**05**
3.	**a.**	Illustrate a table-driven agent function with an example.	L3	CO2	**05**
		OR			
	b.	Choose an appropriate PEAS description for an interactive English tutor agent.	L3	CO2	**05**
4.	**a.**	Explain the problem description of route-finding problem.	L2	CO3	**05**
		OR			
	b.	Outline the pseudocode of simple problem-solving agent.	L2	CO3	**05**
5.	**a.**	Illustrate the properties of Depth First Search.	L3	CO4	**05**
		OR			
	b.	Illustrate the recursive implementation of Depth-limited search.	L3	CO4	**05**
6.	**a.**	Discuss the characterization of Wumpus world.	L2	CO5	**05**
		OR			
	b.	Explain validity and satisfiability.	L2	CO5	**05**
7.	**a.**	Illustrate state space formulation.	L3	CO6	**05**

		OR			
	b.	Illustrate partial order planning.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Explain the gestation of Artificial Intelligence.	L2	CO1	**10**
		OR			
	b.	Discuss the contribution of Computer Science and Engineering to AI.	L2	CO1	**10**
9.	**a.**	Illustrate a utility-based agent with a neat diagram.	L3	CO2	**10**
		OR			
	b.	Sketch in detail the architecture of a learning agent.	L3	CO2	**10**
10.	**a.**	Discuss touring and traveling sales person problem with example.	L2	CO3	**10**
		OR			
	b.	Explain the various phases in problem solving.	L2	CO3	**10**
11.	**a.**	Interpret the performance measures of uninformed search strategies.	L3	CO4	**10**
		OR			
	b.	Demonstrate inventing admissible heuristic functions.	L3	CO4	**10**
12.	**a.**	Outline the syntax of first order logic.	L2	CO5	**10**
		OR			
	b.	Discuss quantification in first order logic.	L2	CO5	**10**
13.	**a.**	Demonstrate heuristic estimation techniques on planning graph.	L3	CO6	**10**
		OR			
	b.	Illustrate partial order planning solution for flat tire problem.	L3	CO6	**10**